

4.3 Sampling and DT Spectra

Purpose of the Experiment

This experiment aims to show the sampling effects on the spectrum of a DT signal. The fast Fourier transform (FFT) algorithm will be implemented by programming and used for analyzing the spectrum of a sampled signal. Based on the experimental results, we can have a good grasp on the physical meaning of the sampling in view of signal reconstruction. Furthermore, as the second best way to avoid the frequency aliasing, the anti-aliasing filter will be taken into account.

Experiment #2.1. Aliasing Effect

Let us examine the effect of frequency aliasing. Theoretically speaking, the aliasing effect occurs whenever a signal is sampled at a rate less than twice the maximum frequency component contained in the given signal. This phenomenon is easily demonstrated using the under-sampled single sinusoid. More complicated situation may be observed if we sample a general signal with a vast number of frequency components.

Now, in order to confirm the aliasing effect, construct the experimental environment as shown in Figure 4.4. For implementing the anti-aliasing filter and the smoothing filter, use the analog circuit depicted in Figure 4.5.

- 1) Using the function generator, generate a sinusoidal waveform $x(t)$ whose magnitude is $4[V_{pp}]$ and frequency is $10[Hz]$. And then, pass it through the anti-aliasing filter.

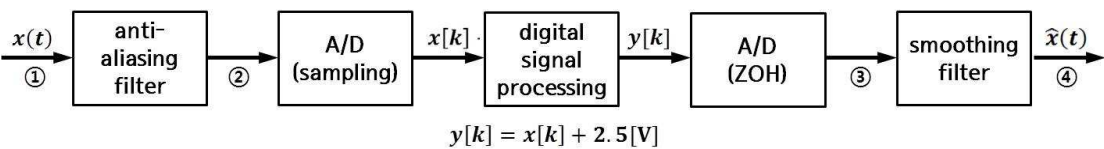


Figure 4.4 Configuration of an experiment to test the frequency aliasing effect

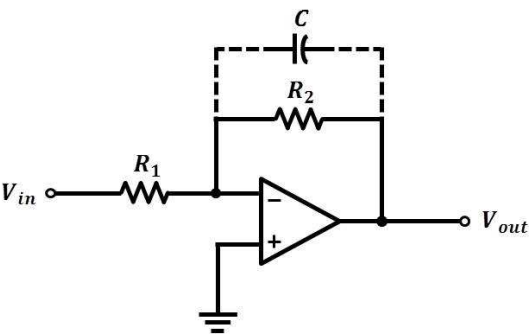


Figure 4.5 Analog filter

- 2) Take out the capacitor from the anti-aliasing filter circuit. That is, the anti-aliasing filter is inactivated in this case. Set the sampling rate as $200[Hz]$ and acquire it from the analog input channel of a DAQ.
- 3) Add the offset voltage $2.5[V]$ to the acquired signal $x[k]$ and output the digital signal $y[k]$ through the analog output channel of a DAQ.
- 4) In order to make the resulting stairwise digital signal $y[k]$ smooth, set the cut-off frequency of the smoothing filter as $30[Hz]$
- 5) Plot the spectra of the signals measured at node ①, ②, ③ and ④. In order to obtain the spectrum of a signal, you can use the FFT function embedded in your oscilloscope. In the spectrum, limit its maximum frequency range to $2f_s$. Observe the frequency component of the signals and analyze your experimental results. What effects do you observe in the waveforms at each node?
- 6) Connect the capacitor to the anti-aliasing filter circuit. This activates the anti-aliasing filter. Set the cut-off frequency of the anti-aliasing filter as $30[Hz]$. Repeat the above experiments from 3)~5).

Experiment #2.2. Fast Fourier Transform and DT Spectra

In this part, we will learn how to use the Matlab embedded FFT function to find the DT spectrum of a discrete-time signal. In addition, we will also make use of the IFFT algorithm in recovering the discrete-time signal from its spectrum.

- 1) Consider the discrete-time signal

$$x[k] = \begin{cases} -1 & , k = 0, 1, 2, 3 \\ 0 & , \text{otherwise} \end{cases}$$

and find its DTFT analytically. Recall the DTFT theory summarized in the lecture note.

- 2) Using the MATLAB function `xw = fft(x, N)`, find the spectrum of the above signal where $N = 2^2, 2^3, 2^4, 2^5$. Also, utilize the MATLAB function `hatx = ifft(xw, N)` to recover the original discrete-time signal from the spectrum `xw`. Plot the magnitude and phase spectrum. As well, plot the original and recovered signals. Briefly comment on the results.
- 3) Implement the 1024-point FFT algorithm as a C function. Solve the problem in 2) using your own C program. It might be helpful to find and look through the reference entitled *Numerical Recipe in C* in your university library together with the lecture note. Verify your C program by comparing the result with that obtained by MATLAB embedded function, `fft`.
- 4) Repeat 1) and 2) for the following discrete-time signal

$$x[k] = \begin{cases} 1 & , k = 0, 1, \dots, 11 \\ 0 & , \text{otherwise} \end{cases}$$